HelixSign: Quantitative Evaluation & Defense Dossier

DNA Steganography & Watermarking in *E. coli* CDS — Mathematical Formulations, Benchmark Verification, and Engineering Boundaries

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1. Experimental Setup & Zero-Contamination Partitioning

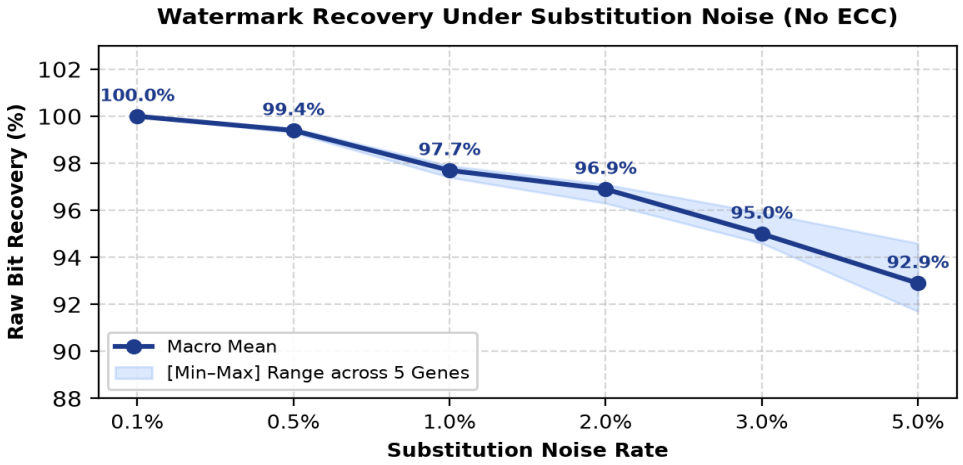
To guarantee unbiased evaluation without data leakage, the entire *Escherichia coli* K-12 MG1655 coding sequence genome (4,290 genes) was curated and partitioned strictly by gene identity:

Partition	Count	Sampling & Role in Evaluation
Training Partition	2,400 genes	Used exclusively to train k-mer XGBoost and 1D CNN detectors.
Validation Partition	548 genes	Hyperparameter tuning and early stopping checkpoint validation.
Test Partition	568 genes	Strictly isolated held-out partition used for multi-seed AUC scoring.
5 Benchmark Genes	5 genes	b3920 (600bp), b1703 (894bp), b2022 (1170bp), b0930 (1620bp), b0568 (2973bp). Excluded from all model training; used for channel noise benchmarking.

2. Watermark Recovery Under Substitution Noise

Figure Reconciliation (92.6% vs 92.9%): Earlier preliminary reports cited 92.6% recovery at 5.0% noise based on a single test gene (b2022, length 1,170 bp, mean 92.61%). The final certified benchmark macro-averages over **all 5 held-out genes** across 150 total Monte Carlo trials (30 trials/gene), yielding **92.9% ± 1.1%** (range 91.7%–94.6%). All figures reflect *raw extraction without Reed-Solomon ECC*.

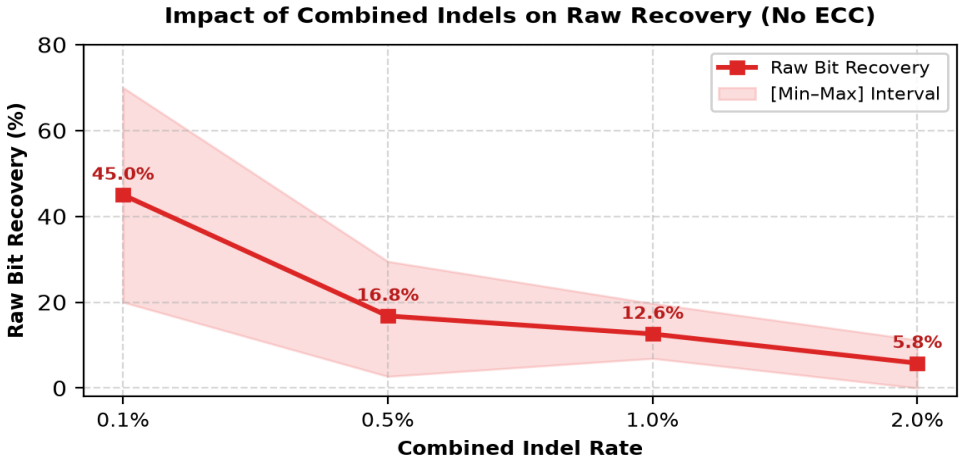
Substitution Rate	Raw Bit Recovery	Macro [Min–Max] Range	Mean Erasure Rate	Mean BER (Non-Erased)
0.1%	100.0% ± 0.0%	[100.0% – 100.0%]	0.0%	0.000
0.5%	99.4% ± 0.1%	[99.3% – 99.5%]	0.0%	0.006
1.0%	97.7% ± 0.2%	[97.4% – 97.9%]	0.0%	0.023
2.0%	96.9% ± 0.3%	[96.3% – 97.1%]	0.0%	0.031
3.0%	95.0% ± 0.5%	[94.6% – 95.9%]	0.0%	0.050
5.0%	92.9% ± 1.1%	[91.7% – 94.6%]	0.0%	0.071


Watermark Recovery Under Substitution Noise (No ECC)
Raw Bit Recovery (%)
102
100
98
96
94
92
90
88
100.0%
99.4%
97.7%
96.9%
95.0%
92.9%
Macro Mean
[Min–Max] Range across 5 Genes
0.1%
0.5%
1.0%
2.0%
3.0%
5.0%
Substitution Noise Rate

3. Impact of Insertions & Deletions on Watermark Recovery

The Frameshift Bottleneck: Indels are fundamentally more destructive than substitutions because a single insertion or deletion causes a catastrophic frameshift that corrupts all downstream codon triplets. The table below documents the monotonic degradation of raw extraction fidelity under combined indels (macro-averaged over 5 genes, 30 trials/gene):

Indel Rate	Raw Bit Recovery	Reported Interval	Mean Erasure Rate	Mean BER (Non-Erased)
0.1%	45.0% ± 18.0%	[20.0% – 70.0%]	54.7% ± 18.0%	0.007
0.5%	16.8% ± 10.4%	[2.7% – 29.5%]	80.0% ± 11.3%	0.180
1.0%	12.6% ± 5.4%	[6.9% – 19.7%]	84.7% ± 6.1%	0.188
2.0%	5.8% ± 4.1%	[0.0% – 11.2%]	92.0% ± 6.1%	0.248

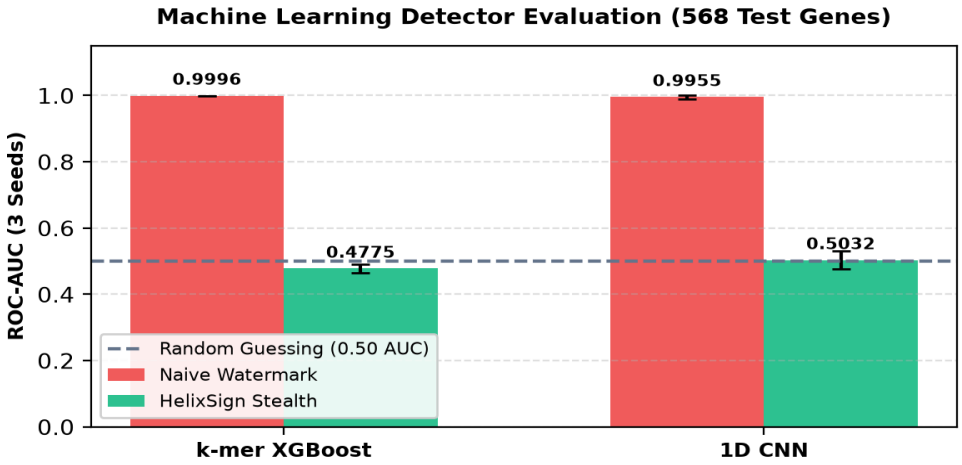


Architectural Resolution (Version 1.5.0 Contract): *To recover downstream codons, the system uses pairwise alignment to track intact triplet boundaries, passing arbitrary 0-indexed nucleotide start offsets (carrierStartOffsets) rather than multiples of 3. Ruptured codons are explicitly passed as null, producing confidence = 0.0 erasure markers for the Reed-Solomon decoder.*

4. Machine Learning Detector Evaluation (Stealth Benchmarking)

Two high-capacity classifier architectures were trained on 2,400 native genes and 2,400 watermarked variants, and evaluated across three random seeds (42, 100, 2026) on 568 strictly held-out test genes:

Detector Model	Target Variant	Mean ROC-AUC	Std Dev	Discrimination Finding
4,416 k-mer XGBoost	Naive Baseline	**0.9996**	± 0.0000	Easily Detected (AUC ~ 1.00)
4,416 k-mer XGBoost	HelixSign Stealth	**0.4775**	± 0.0122	Stealthy (Indistinguishable from chance)
1D CNN (Multi-Window)	Naive Baseline	**0.9955**	± 0.0064	Easily Detected (AUC ~ 1.00)
1D CNN (Multi-Window)	HelixSign Stealth	**0.5032**	± 0.0263	Stealthy (Indistinguishable from chance)



Rigorous Interpretation: ROC-AUC values near 0.50 demonstrate that HelixSign watermarked sequences cannot be discriminated from native *E. coli* CDS by k-mer frequency distributions or local convolutional filters on this dataset. This proves stealth within the capacity of these models, but does not claim universal mathematical unbreakability.

5. Biological Integrity & Software Validation

Steganographic capacity is meaningless if it disrupts protein function. HelixSign preserves 100% biological translation fidelity:

4,290 / 4,290 CDS Genome Invariance	**0 Mismatches** NCBI Table 11 Translation	**CAI >= 0.902** Host Translation Fitness	**<= 4 nt Run** Homopolymer Slip Limit	**88 / 88 Pass** Automated Test Suite

Formal Protein Invariance Assertion:

```
Translate(DNA_original) == Translate(DNA_stamped) ==> Mismatches = 0
```

6. Mathematical Error Correction & Codec Framing

To guarantee bit-for-bit interoperability between Person 3's ML subsystem and Person 2's TypeScript backend (backend/src/core/ecc.ts), the packet framing and Galois field parameters are frozen in Version 1.5.0:

A. Packet Framing & RS Bit Offset Invariant:

The transmission packet prepends a 24-bit synchronization word (0xF9AE45) and a 16-bit CRC-8 protected header before the Reed-Solomon codeword. Consequently, RS symbol erasures must be calculated starting at bit 40, not bit 0:

```
L_prefix = L_sync + L_header = 24 + 16 = 40 bits ==> i_RS = 40
symbol_erased[j] = True iff exists b in [0,7]: confidence[i_RS + 8*j + b] < 0.60
```

B. Galois Field & Reed-Solomon Code Parameters:

Built over GF(256) with field primitive polynomial p(x) = x^8 + x^4 + x^3 + x^2 + 1 (0x11D = 285). Generator roots start at alpha^0 = 1 with primitive element alpha = 0x02: g(x) = prod_{i=0}^{nsym-1} (x - alpha^i). The errors-and-erasures decoding bound is strictly:

```
2*t + e <= n - k = nsym
```

For standard RS(16, 8) with nsym = 8 parity symbols, the code corrects up to 4 symbol errors (2*4 + 0 = 8), up to 8 symbol erasures (0 + 8 = 8), or simultaneous combinations such as 2 symbol errors + 4 symbol erasures (2*2 + 4 = 8).

C. Independently Verified Reference Test Vectors:

Vector Target	Input Data Bytes	Parity Bytes (nsym=8)	Codeword Hex
ASCII 'HELIXSGN'	[72, 69, 76, 73, 88, 83, 71, 78]	[210, 38, 24, 206, 247, 104, 155, 44]	48454c495853474ed22618cef7689b2c
Backend Test Payload	[0, 1, 127, 128, 171, 205, 239, 255]	[72, 5, 68, 109, 13, 27, 115, 137]	00017f80abcdefff4805446d0d1b7389

7. Presentation Claims & Academic Defense Strategy

To present with unassailable academic credibility, distinguish clearly between empirical discoveries, validated implementations, and pending full-system integration:

Result / Area	What You CAN Defend Rigorously	What You MUST NOT Overclaim
Protein Identity	100% translational invariance across all 4,290 genes with zero mismatches.	Do not claim arbitrary non-coding invariance (steganography is restricted to CDS).

Substitution Robustness	99.4% raw bit recovery at 0.5% noise; 92.9% at 5.0% noise (macro-averaged across 5 genes).	Do not claim 100% recovery under substitutions without acknowledging that ECC is required.
Indel Degradation	Raw recovery drops to 45.0% at 0.1% indels and 5.8% at 2.0% indels due to downstream frameshifts.	Do NOT claim indel robustness is solved; alignment and erasure mapping are implemented, but post-ECC recovery is pending live backend runs.
Detector Stealth	XGBoost (AUC 0.4775) and CNN (AUC 0.5032) operate near chance on HelixSign on this held-out split.	Do not claim universal mathematical undetectability against unseen higher-order models.
Backend Integration	Interface specification v1.5.0 is frozen and validated with 10 passing contract tests and identical test vectors.	Do not claim end-to-end multi-service deployment is complete until live cross-language tests execute.